

CIS 121 Introduction to Programming

Problem Set – Nested If and compound relational conditions. For each problem create an IPO chart and c++ code file.

1. The input to the problem is quantity of widgets and customer status. You set the price based upon quantity and status using the table below. Your program should determine the price to charge based on the schedule below. Calculate the extended price. Calculate tax at 7%. Display the extended price, tax amount and total.

Quantity	Status	Price
>10000	A	\$10
>10000	B	\$12
5000 to 10000	C	\$20
5000 to 10000	D	\$22
Below 5000	Any	\$30

Input	Process	Output
- Quantity of Widgets - Customer Status	- Assign a variable to the input for Quantity of Widgets - Assign a variable to the input for Customer Status - Declare Price and set at \$10 if Quantity is greater than 10,000 and Status equals A - If Quantity is greater than 10,000 and Status equals B, set price at \$12 - If Quantity is greater than or equal to 5,000 and Status equals C, set price at \$20 - If Quantity is greater than or equal to 5,000 and Status equals D, set price at \$22 - If Quantity is less than 5,000 and Status equals either A, B, C, or D, set price at \$30 - Otherwise, End program - Calculate Extended Price by finding the product of Quantity and Price - Calculate Tax by finding the product of Extended Price and 7% (.07) - Calculate Total by finding the sum of Extended Price and Tax - Display Extended Price, Tax, and Total	- Extended Price - Tax Amount - Total

2. Enter a part number of the following (10, 99, 55, 70, 50). Also enter the quantity. Determine the cost per unit using the table below. Then calculate the total cost. Display the part number, cost per unit and total.

Part	Quantity	Cost Per Unit
10	> 1000	1.00
99	> 500	2.00
All others	All others	5.00

Input	Process	Output
- Part Number - Quantity	- Assign a variable to the input for Part Number - Assign a variable to the input for Quantity - Declare Cost/Unit and set at \$1.00 if Part Number equals 10 and Quantity is greater than 1,000 - If Part Number equals 99 and Quantity is greater than 500, set Cost/Unit at \$2.00 - If Part Number equals either 55, 70, or 50, set Cost/Unit at \$5.00 - Otherwise, End program - Calculate Total by finding the product of Quantity and Cost/Unit - Display Part Number, Cost/Unit, and Total	- Part Number - Cost per Unit - Total

3. Allow the user to enter number of concert tickets and location code (H, L). The price per ticket depends on the volume and location (see below). Display the number of tickets, price per ticket and the total cost.

Volume is greater than 25 or location is H cost per ticket is \$30.00

Volume is greater than 10 (10 to 24) or location is L cost per ticket is \$40.00

All other quantities or locations are \$50.00

Input	Process	Output
- Quantity of Concert Tickets - Location code	- Assign a variable to the input for Quantity - Assign a variable to the input Location Code - Declare Price/Ticket and set at \$30.00 if Quantity is greater than or equal to 25 or Location Code equals H - If Quantity is greater than 10 or Location Code equals L, set Price /Ticket at \$40.00	- Quantity of Concert Tickets - Price per Ticket - Total Cost

	○ Otherwise, set Price /Ticket at \$50.00 ● Calculate Total Cost by finding the product Quantity and Cost/Ticket ● Display Quantity, Price /Ticket, and Total Cost	
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4. Allow the user to enter equipment code of a rental and a code indicating half day or full day. Determine the cost of the rental. Display the rental cost

Equipment Code	Day	Cost
A	F	10.00
A	H	15.00
B	F	20.00
B	H	35.00
C	H	40.00
C	F	45.00
All others	All others	50.00

Input	Process	Output
● Equipment Code ● Duration Code	● Assign a variable to the input for Equipment Code ● Assign a variable to the input for Duration Code ● Declare Cost and set at \$10.00 if Equipment Code equals A and Duration Code equal Full-day ○ If Equipment Code equals A and Duration Code equals Half-day, set Cost at \$15.00 ○ If Equipment Code equals B and Duration Code equals Full-day, set Cost at \$20.00 ○ If Equipment Code equals B and Duration Code equals Half-day, set Cost at \$35.00 ○ If Equipment Code equals B and Duration Code equals Full-day, set Cost at \$20.00 ○ If Equipment Code equals C and Duration Code equals Half-day, set Cost at \$40.00 ○ If Equipment Code equals C and Duration Code equals Full-day, set Cost at \$45.00 ○ Otherwise, set Cost at \$50.00 ● Display Cost	● Rental Cost

5. You need to display the gross salary for an employee. They input a job code and Hours. First, determine the rate of pay based on job code and hours (see table below). Next, compute gross pay (hrs * rate). No overtime pay.

Job Code	Hours	Rate of Pay
L	> 40	50.00
L	<=40	40.00
J	>60	100.00
J	<=60	75.00
A	>40	25.00
A	<=40	20.00

Input	Process	Output
- Job Code - Hours	- Assign a variable to the input for Job Code - Assign a variable to the input for Hours - Declare Rate of Pay and set at \$50.00 if Job Code equals L and Hours are greater than 40 - If Job Code equals L and Hours are less than or equal to 40, set Rate of Pay at \$40.00 - If Job Code equals J and Hours are greater than 60, set Rate of Pay at \$100.00 - If Job Code equals J and Hours are less than or equal to 60, set Rate of Pay at \$75.00 - If Job Code equals A and Hours are is greater than 40, set Rate of Pay at \$25.00 - If Job Code equals A and Hours are less than or equal to 40, set Rate of Pay at \$20 - Otherwise, End Program - Calculate Gross Pay by finding the product of Hours and Rate of Pay - Display Gross Pay	Gross Pay